

Module 4- Data Mining Techniques

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MODULE - IV

Data Mining Techniques: Association Rule Mining: Efficient and Scalable Frequent Item set Mining Methods, Mining, Various Kinds of Association Rules, Association Rules, Market Basket Analysis, Apriori Algorithm, Ethical Issues in Data Science and Data Mining.

10 Hours

Data Mining Techniques

Data Mining is the process of discovering useful **patterns, trends, and insights** from **large sets of data**.

It helps in making decisions, predictions, and understanding hidden relationships that are not obvious.

Data Mining Techniques

- **Frequent patterns** are patterns (e.g., itemsets, subsequences, or substructures) that **appear frequently** in a data set.
- For example, a set of items, such as milk and bread, that appear frequently together in a transaction data set is a **frequent itemset**.
- A subsequence, such as buying first a PC, then a digital camera, and then a memory card, if it occurs frequently in a shopping history database, is a **(frequent) sequential pattern**.

Data Mining Techniques

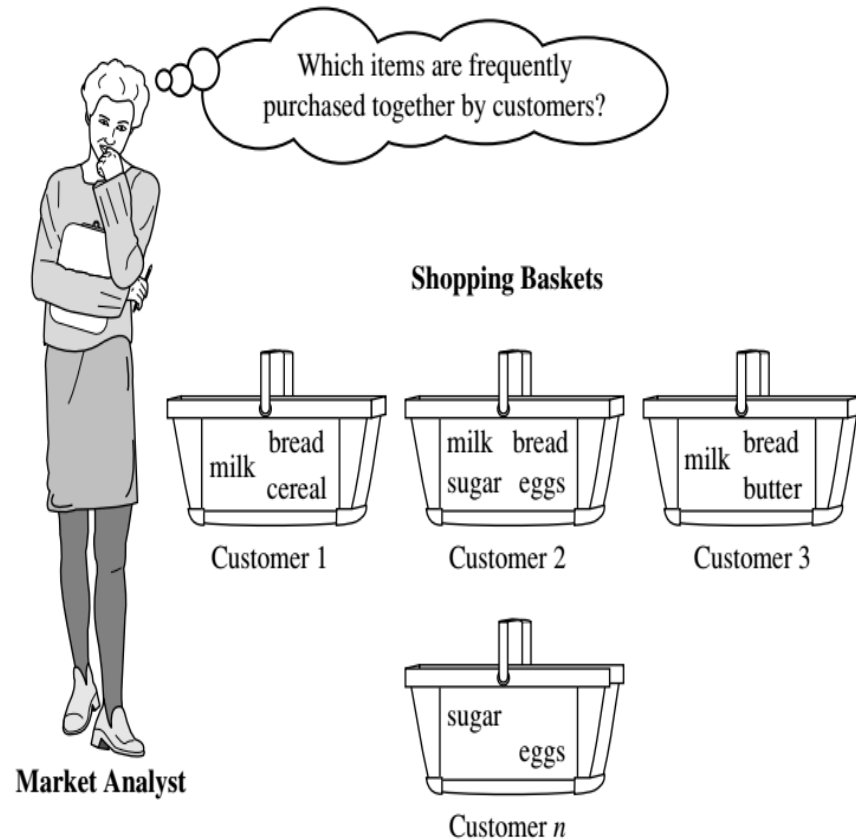
- **Finding frequent patterns** plays an essential role in mining associations, correlations, and many other interesting relationships among data.
- Moreover, it helps in data classification, clustering, and other data mining tasks. Thus, frequent pattern mining has become an important data mining task and a focused theme in data mining research.
- **Note:** A recommendation system can be built using *frequent pattern mining* (especially using association rules), but not all recommendation systems use frequent pattern mining.

Data Mining Techniques

- Frequent pattern mining searches for recurring relationships in a given data set. i.e to discover the interesting associations and correlations between itemsets in transactional and relational databases.
- **Market Basket Analysis (MBA)**, is the earliest form of an example for **frequent pattern mining (frequent itemset mining)** for association rules.
- Market basket analysis uses data mining technique that uncovers **co-purchasing patterns**.
- This process analyzes customer buying habits by finding associations between the different items that customers place in their “shopping baskets” as shown in figure (next slide)...

Data Mining Techniques

WHY DO WE NEED MARKET BASKET ANALYSIS (MBA)?



- To understand customer buying behaviour.
- To optimize store layout.
- Cross-selling & up-selling.
- Promotions & discounts.
- Personalized recommendation engines.

Figure 6.1 Market basket analysis.

Data Mining Techniques: Association Rule Mining

Association Rule Mining (ARM) is a **data mining technique** used to discover *relationships, patterns, or associations* between items in large datasets.

It helps answer questions like:

- “Which items are frequently bought together?”
- “If a customer buys **A**, how likely are they to buy **B**?”

Association Rule Mining finds **if–then** patterns in data.

Data Mining Techniques: Association Rule Mining (ARM)

Structure of an Association Rule:

A rule has two parts:

1. Antecedent (IF part)

- Items or conditions that appear first.

2. Consequent (THEN part)

- Items likely to appear along with antecedent.
- Example:

IF (Tea, Sugar) THEN (Milk)

Data Mining Techniques: Association Rule Mining

Measures Used:

To check the strength of rules, ARM uses:

1. Support

- How often items appear together in the dataset (Frequency of an itemset).

2. Confidence

- How often the rule is true (Strength of implication).

3. Lift

- How much more likely the items occur together **compared to chance** (Strength of association compared to random chance).

Data Mining Techniques: Association Rule Mining

- **Association Rule Mining (ARM)** is applied in Market Basket Analysis to analyze the data to learn about the purchasing behavior of their customers.
- Besides market basket data, association analysis is also applicable to data from other application domains such as **bioinformatics, medical diagnosis, web mining, and scientific data analysis.**
- In the **analysis of Earth science data**, for example, association patterns may reveal interesting connections among the ocean, land, and atmospheric processes.

Data Mining Techniques: Association Rule Mining

How Association rule mining is applied to Market Basket Analysis?

Huge amounts of customer purchase data are collected daily at the checkout counters of grocery stores. Table 4.1 shows market basket transactions.

Each row in this table corresponds to a transaction, which contains a unique identifier labelled TID and a set of items bought by a given customer.

Retailers are interested in analysing the data to learn about the purchasing behaviour of their customers.

Table 4.1. An example of market basket transactions.

<i>TID</i>	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}

Data Mining Techniques: Association Rule Mining

The **uncovered relationships** can be represented in the form of sets of items present in many transactions, which are known as **frequent itemsets**, or **association rules**, that represent relationships between two itemsets.

STEP 1: Collect Transaction Data

Table 4.1. An example of market basket transactions.

<i>TID</i>	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}

Data Mining Techniques: Association Rule Mining

STEP 2: **Convert to Binary (0/1) Table** (because Simplifies computation, Enables mathematical operations, Efficient storage and access) :

Market basket data can be represented in a **binary format** as shown in Table 4.2, where each row corresponds to a transaction and each column corresponds to an item.

An item can be treated as a binary variable whose value is **one** if the item is **present** in a transaction and **zero** otherwise.

Data Mining Techniques: Association Rule Mining

Binary table (1 = purchased, 0 = not purchased)

Table 4.1. An example of market basket transactions.

<i>TID</i>	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}

Table 4.2. A binary 0/1 representation of market basket data.

TID	Bread	Milk	Diapers	Beer	Eggs	Cola
1	1	1	0	0	0	0
2	1	0	1	1	1	0
3	0	1	1	1	0	1
4	1	1	1	1	0	0
5	1	1	1	0	0	1

Data Mining Techniques: Association Rule Mining

Step 3: Find Frequent Itemsets (Co-purchased Groups)- i.e. Find the Frequency of Each Item (Support) : i.e. Count occurrences of each item:

Item	Frequency	Frequent?
Bread	4	✓
Milk	4	✓
Diapers	4	✓
Beer	3	✓
Cola	2	✓
Eggs	1	✗ (Discard)

Then apply **minimum support threshold=2**. (e.g., **2** transactions). So, **items with support < 2 are removed. Eggs is removed. This produces L1 (frequent 1-itemsets).**

L1 = {Bread, Milk, Diapers, Beer, Cola}

Data Mining Techniques: Association Rule Mining

Step 4: Generate Candidate 2-Itemsets (C₂)

Combine all L₁ items in all possible pairs: Using your item list from **L₁ (frequent 1-itemsets)**: **L₁ = {Bread, Milk, Diapers, Beer, Cola}**

- i. {Bread–Milk}
- ii. {Bread–Diapers}
- iii. {Bread–Beer}
- iv. {Bread–Cola}
- v. {Milk–Diapers}
- vi. {Milk–Beer}
- vii. {Milk–Cola}
- viii. {Diapers–Beer}
- ix. {Diapers–Cola}
- x. {Beer–Cola}

Number of pairs = $5C_2 = 10$.

Data Mining Techniques: Association Rule Mining

Compute support (how many transactions contain each pair): Support Count Table for All C2 Itemsets.

Table 4.1. An example of market basket transactions.

<i>TID</i>	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}

C2 Itemset	Appears In Transactions	Support Count
{Bread, Milk}	T1, T4, T5	3
{Bread, Diapers}	T2, T4, T5	3
{Bread, Beer}	T2, T4	2
{Bread, Cola}	T5	1
{Milk, Diapers}	T3, T4, T5	3
{Milk, Beer}	T3, T4	2
{Milk, Cola}	T3, T5	2
{Diapers, Beer}	T2, T3, T4	3
{Diapers, Cola}	T3, T5	2
{Beer, Cola}	T3	1

Data Mining Techniques: Association Rule Mining

Again, apply **min-support = 2** $\rightarrow$ keep pairs with count ≥ 2 .

Itemset	Support	Frequent?
Bread, Milk	3	✓
Bread, Diapers	3	✓
Bread, Beer	2	✓
Bread, Cola	1	✗
Milk, Diapers	3	✓
Milk, Beer	2	✓
Milk, Cola	2	✓
Diapers, Beer	3	✓
Diapers, Cola	2	✓
Beer, Cola	1	✗

Hence, **L2 = all pairs except {Bread, Cola} and {Beer, Cola}**

Data Mining Techniques: Association Rule Mining

Step 5: Generate Candidate 3-Itemsets (C₃)

Rule:

- You can only combine itemsets that share prefix items (i.e. if they share **1 common item**).

Example:

(i) {**Bread**–Milk} and {**Bread**–Diapers} share “**Bread**”, so join them:

{Bread, Milk, Diapers}

(ii) {**Bread**, Milk} and {**Bread**, Beer} : **{Bread, Milk, Beer}**

Data Mining Techniques: Association Rule Mining

- (iii) {**Bread**, Diapers} and {**Bread**, Beer}: {**Bread, Diapers, Beer**}
- (iv) {**Milk**, Diapers} and {**Milk**, Beer} : {**Milk, Diapers, Beer**}
- (v) {**Milk**, Diapers} and {**Milk**, Cola} : {**Milk, Diapers, Cola**}
- (vi) {**Milk**, Beer} and {**Milk**, Cola}: {**Milk, Beer, Cola**}
- (vii) {**Diapers**, Beer} and {**Diapers**, Cola} : {**Diapers, Beer, Cola**}

Total 7 candidates are there.

Table 4.1. An example of market basket transactions.

TID	Items
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5	{Bread, Milk, Diapers, Cola}

{Milk, Diapers, Cola}

Subsets:

- {Milk, Diapers} ✓
 - {Milk, Cola} ✓
 - {Diapers, Cola} ✓
- **KEEP**

{Bread, Milk, Diapers}

• Subsets:

- {Bread, Milk} ✓
 - {Bread, Diapers} ✓
 - {Milk, Diapers} ✓
- **KEEP**

{Bread, Milk, Beer}

• Subsets:

- {Bread, Beer} ✓
 - {Bread, Milk} ✓
 - {Milk, Beer} ✓
- ? **REMOVE**

{Bread, Diapers, Beer}

Subsets:

- {Bread, Diapers} ✓
 - {Bread, Beer} ✓
 - {Diapers, Beer} ✓
- **KEEP**

{Milk, Diapers, Beer}

Subsets:

- {Milk, Diapers} ✓
 - {Milk, Beer} ✓
 - {Diapers, Beer} ✓
- **KEEP**

{Milk, Beer, Cola}

- Subsets:
- {Milk, Beer} ✓
- {Milk, Cola} ✓
- {Beer, Cola} ✗ (support = 1 only, NOT in L2)
- → ? REMOVE

{Diapers, Beer, Cola}

Subsets:

- {Diapers, Beer} ✓
 - {Diapers, Cola} ✓
 - {Beer, Cola} ✗ (not frequent)
- ? REMOVE

Itemset	Support	Frequent?
Bread, Milk, Diapers	2	✓
Bread, Diapers, Beer	2	✓
Milk, Diapers, Beer	2	✓
Milk, Diapers, Cola	2	✓

Hence, L3 = all above 4-itemsets

Data Mining Techniques: Association Rule Mining

Step 6: Generate Candidate 4-Itemsets (C₄)

Example combination 1:

{Bread, Milk, Diaper} and **{Bread, Diapers, Beer}** gives rise to
{Bread, Milk, Diapers, Beer}

Check the support value of this 4-pair items:

{Bread, Milk, Diapers, Beer} Appears only in T₄ ⇒ **support = 1**

Hence, Not frequent ⇒ DISCARDED

So, no 4-itemsets are frequent.

FINAL FREQUENT ITEMSETS (F)

These are ALL frequent patterns:

1-itemsets: Bread, Milk, Diapers, Beer, Cola

2-itemsets: Bread-Milk, Bread-Diapers, Milk-Diapers, etc.

3-itemsets: Bread-Milk-Diapers, Milk-Diapers-Beer, etc.

4-itemsets: None

Data Mining Techniques: Association Rule Mining

STEP 7: Generate Association Rules from Frequent Itemsets

- How to split frequent itemsets into left (X) and right (Y)
- How to evaluate rules using metrics

STEP 7.1 — Generating Rules from a 3-itemset

Take a frequent itemset: {Bread, Milk, Diapers}

All possible rules are:

- Bread, Milk $\rightarrow$ Diapers
- Bread, Diapers $\rightarrow$ Milk
- Milk, Diapers $\rightarrow$ Bread
- Bread $\rightarrow$ Milk, Diapers
- Milk $\rightarrow$ Bread, Diapers
- Diapers $\rightarrow$ Bread, Milk

Now we compute **confidence**, **lift**, etc.

Data Mining Techniques: Association Rule Mining

STEP 7.2 — Evaluate Rule using METRICS

Metrics tell us whether a rule is meaningful.

- {Bread, Milk, Diapers} → Appears in T4 and T5

1. **SUPPORT:** measures how common is this pattern in the entire dataset

$$\text{Support}(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{N}$$

Example:

Itemset = {Bread, Milk, Diapers}

It appears in **2** of 5 transactions.

$$\text{Support} = \frac{2}{5} = 0.4 = 40\%$$

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5	{Bread, Milk, Diapers, Cola}

Support = 2 transactions = 2 transactions out of 5 transactions = 2/5=0.4= 40%

“Support tells us how often the pattern occurs in the dataset.

A support of 40% means this combination happened in 40% of all shopping baskets.”

Data Mining Techniques: Association Rule Mining

2. CONFIDENCE: measures how often the rule is correct.

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Transactions containing both X and Y}}{\text{Transactions containing X}}$$

i.e.
$$\text{Confidence}(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)}$$

Example Rule:

$\{\text{Bread, Milk}\} \rightarrow \{\text{Diapers}\}$

We need:

- $\sigma(\text{Bread \& Milk \& Diapers}) = 2$
- $\sigma(\text{Bread \& Milk}) = 3$

$$\text{Confidence} = \frac{2}{3} = 0.66 = 66\%$$

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Confidence tells $\rightarrow$ “If 100 customers buy Bread and Milk together, 66 of them ALSO buy Diapers.
Confidence measures the rule’s accuracy.”

Data Mining Techniques: Association Rule Mining

3. LIFT: Is the rule truly strong or just a coincidence?

- Lift compares: Confidence **against** How popular the consequent (Y) already is ?

$$\text{Lift} = \frac{\text{Confidence}(X \rightarrow Y)}{\text{Support}(Y)}$$

Example:

- Rule = Bread, Milk $\rightarrow$ Diapers
- Confidence = 0.66
- Support(Diapers) = 4/5 = 0.8

$$\text{Lift} = \frac{0.66}{0.8} = 0.825$$

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“Lift checks whether X really helps predict Y,
or Y is just popular anyway.”

Data Mining Techniques: Association Rule Mining

Step 8: Interpret Rules

- **Lift** > 1 $\rightarrow$ Strong positive association
- **Lift** < 1 $\rightarrow$ Items are negatively associated
- **Lift** $= 1$ $\rightarrow$ Purely independent

For {Bread, Milk, Diapers}, **Lift** $< 1 \rightarrow$ **Diapers are very common, so the rule is not special**

Step 9: Make business decisions

- To optimize store layout.
- Cross-selling & up-selling.
- Promotions & discounts.

*****END OF PROBLEM SOLVING*****

Solve few examples

Example 1:

T1: {Burger, Fries, Coke}

T2: {Burger, Fries}

T3: {Coffee, Donut}

T4: {Pizza, GarlicBread}

T5: {Burger, Coke}

T6: {Pizza, Coke}

T7: {Coffee}

T8: {Burger, Fries, Donut}

T9: {Pizza, GarlicBread, Coke}

T10: {Coffee, Donut, Coke}

Calculate Metrics, interpret rules for below rule pairs and make business decisions

Rule pair: **Pizza → Garlic Bread**

Rule pair: **Burger → Donut**

Rule pair: **Coffee → Coke**

Example 2:

T1: {Laptop, Mouse, Bag}

T2: {Router, Cable}

T3: {Laptop, Mouse}

T4: {Phone, Charger}

T5: {Laptop, Headphones}

T6: {Router, ExtensionCord}

T7: {Phone, ScreenGuard}

T8: {Laptop, Bag}

T9: {Headphones, Mouse}

T10: {Phone, Charger, Earbuds}

TYPES OF PATTERNS FOUND IN MBA:

✓ **Complementary patterns**

{Chips → Cold Drink}

✓ **Substitutes**

{Tea → Coffee (Negative lift)}

✓ **Seasonal patterns**

{Umbrella → Raincoat}

✓ **High-value bundles**

{Laptop → Mouse → Antivirus}

Summary of MBA

- Market Basket Analysis starts with recording all customer purchases.
- Every shopping bill becomes a **transaction**.
- These transactions are converted into a simple **matrix** of 0s and 1s.
- Algorithms then scan this matrix to find items that occur together often. These groups of items are called **frequent itemsets**.
- From frequent itemsets we generate association rules like ‘**If someone buys X, they are likely to buy Y.**’
- We **evaluate** these rules using support, confidence, and lift.
- Finally, the **store uses these patterns** for product placement, discounts, and recommendations.
- This entire flow — from raw data to business action — is “Market Basket Analysis.”

APRIORI ALGORITHM

Purpose

- Apriori finds **frequent itemsets** from a transaction database, which are later used to generate **association rules** (like market basket patterns).

Inputs to the Algorithm

- Transaction database, minimum support threshold (min_sup)

Principle

- If an itemset is frequent, all its subsets must also be frequent (Downward Closure Property).

i.e. If a combination of items — say $\{A, B, C\}$ — appears frequently in the dataset, then every smaller part of it, $\{A, B\}$, $\{A, C\}$, $\{B, C\}$, $\{A\}$, $\{B\}$, $\{C\}$ must also appear frequently.

- This property helps eliminate large numbers of impossible candidates early (**pruning**)

Working of the Algorithm

From the Transaction Database:

STEP 1 — Form L1 (Frequent 1-itemsets)

- L1 is formed directly from counting individual items.
- Apply min_support.

STEP 2 — Generate C2 (Candidate 2-itemsets)

- Use L1 to generate C2 through JOIN.

Example: if $L1 = \{A, B, C\}$, $C2 = \{AB, AC, BC\}$

- Now we have the first candidate set, which is C2.

STEP 3 — Prune C2

- Remove any candidate pair where one of its subsets is not in L1.

Check subsets of each pair:

- $AB \rightarrow A, B \rightarrow$ both in L1 ✓
- $AC \rightarrow A, C \rightarrow$ both in L1 ✓
- $BC \rightarrow B, C \rightarrow$ both in L1 ✓

No candidate fails pruning. So, all stay in C2.

STEP 4 — Count support for remaining C2

- Scan the transaction, compute support.

STEP 5 — Form L2

- Keep only those candidates whose support $\geq$ min_support.

Then repeat the pattern:

To get L3:

- Join L2 $\rightarrow$ get C3
- Prune C3 (Remove any candidate pair where one of its subsets is not in L2).
- Count support
- Form L3

To get L4:

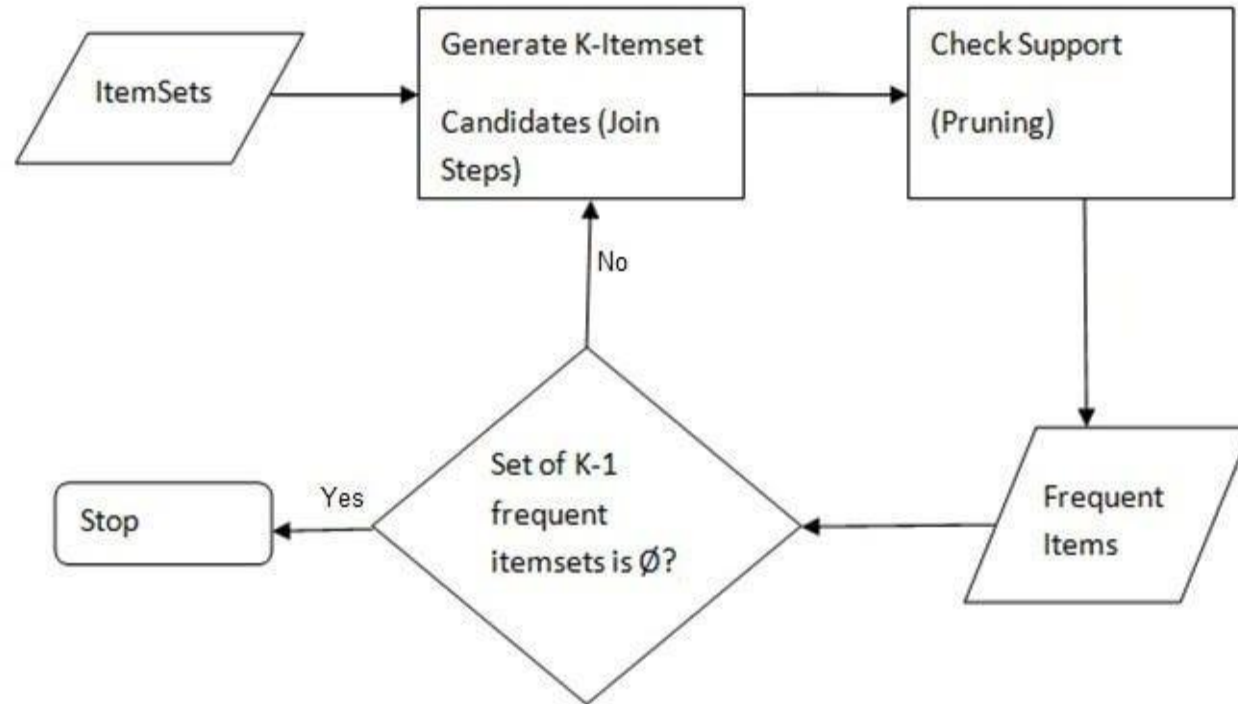
Join L3 $\rightarrow$ get C4

- Prune C4 (Remove any candidate pair where one of its subsets is not in L3).
- Count support
- Form L4

... and so on until no more itemsets.

Note: pruning matters most at C3 or C4 or ... C_k in the standard algorithm.

Database $\rightarrow$ L1 $\rightarrow$ C2 $\rightarrow$ L2 $\rightarrow$ C3 $\rightarrow$ L3 $\rightarrow$ C4 $\rightarrow$ L4 $\rightarrow$...



Pruning Example

Use `min_support = 2`

T1: A B

T2: B C

T3: A B

T4: B C

T5: A D

T6: B D

`Items`: A, B, C, D.

`Total transactions` = 6.

Note:

When forming C3 (or higher), we check each candidate's (k-1)-subsets; if any subset is not in L(k-1), we prune the candidate immediately — without counting it and save time.

❑ Step 1 — Form L1

All counts $\geq \text{min_sup}$ (2),

`L1` = {A, B, C, D}

❑ Step 2 — Generate C2

`C2` = {AB, AC, AD, BC, BD, CD}

Here, we form C2 them from L1, so their single-element subsets are known frequent items — hence pruning C2 is unnecessary

❑ Step 3 — Count support for each C2

AB $\rightarrow$ count = 2, AC $\rightarrow$ count = 0, AD $\rightarrow$ count = 1, BC $\rightarrow$ count = 2, BD $\rightarrow$ count = 1 and CD $\rightarrow$ count = 0

❑ Step 4 — Form L2

`L2` = {AB, BC} — keep only pairs with count ≥ 2 .

Now, iterate above pattern.

❑ Step 5 — Generate C3

From L2, both AB and BC share B in middle; the union gives,

`C3` = {ABC}

❑ Step 6 — PRUNE C3

Let us examine subsets of candidate ABC:

AB — is AB in L2? $\rightarrow$ Yes (count 2).

AC — is AC in L2? $\rightarrow$ No (AC had count 0 and is not in L2).

BC — is BC in L2? $\rightarrow$ Yes (count 2).

Because AC is not frequent, Apriori PRUNES candidate ABC immediately — we do not need to count support(ABC) in the database.

Outputs of Apriori

- Apriori produces **frequent itemsets**: $L_1, L_2, L_3, \dots, L_k$
- These itemsets are input for rule generation.

Advantages & Limitations

Advantages:

- Simple
- Easy to implement
- Good for small–medium datasets

Limitations:

- Too many candidates
- Many database scans
- Slow on large datasets

Apply Apriori algorithm on below transaction data

- T1: A B C
- T2: A C D
- T3: B C E
- T4: A B C E
- T5: B D E
- T6: A B D
- T7: A C E
- T8: B C D
- T9: A B C D E
- T10: C D E

Find L1 (frequent 1-itemsets)

Item	TIDs	Support
A	T1, T2, T4, T6, T7, T9	6
B	T1, T3, T4, T5, T6, T8, T9	7
C	T1, T2, T3, T4, T7, T8, T9, T10	8
D	T2, T5, T6, T8, T9, T10	6
E	T3, T4, T5, T7, T9, T10	6

All items meet min-support $\rightarrow$ **L1** = {A, B, C, D, E}

During **L1 generation**, if you choose **min_sup = 3**, any item that appears **less than 3 times** is discarded.

How do we decide min_sup?

There is no universal rule, but we use practical guidelines depending on:

(a) Size of the dataset

- For small datasets (like 10–50 transactions):
→ min_sup is usually 2 or 3
- For **large datasets** (like 100K+ transactions):
→ min_sup might be **0.1%, 1%, 5%** depending on the use case.

(b) Nature of the application

Application	Typical min_sup	Why?
Supermarket Market Basket	Low (0.1–1%)	Rare combinations may still matter
Healthcare diagnosis	Medium to high (5–20%)	Only strong, common patterns matter
Fraud detection	Very low (0.01–0.1%)	Rare patterns are important

Note: Domain knowledge is required

How many patterns you want

- High min_sup $\Rightarrow$ **fewer patterns**, faster computation
- Low min_sup $\Rightarrow$ **many patterns**, slower, may explode exponentially

How we chose min_sup in our example

We used **10 transactions**, so good choices are:

- If min_sup = 1 → everything becomes frequent → no use
- If min_sup = 2 → too many patterns
- If min_sup = 3 → balanced
- If min_sup = 4 or 5 → too strict, many useful patterns disappear

So, we selected: **min_sup = 3**

Because:

- It removes weak/rare items (those that appear only 1–2 times)
- It gives enough frequent itemsets to illustrate the Apriori steps
- It avoids the explosion of too many candidates

Generate C2 and form L2

C2 — Candidate 2-itemsets (all combinations from L1) with TIDs & support

Candidate	TIDs	Support
AB	T1, T4, T6, T9	4
AC	T1, T2, T4, T7, T9	5
AD	T2, T6, T9	3
AE	T4, T7, T9	3
BC	T1, T3, T4, T8, T9	5
BD	T5, T6, T8, T9	4
BE	T3, T4, T5, T9	4
CD	T2, T8, T9, T10	4
CE	T3, T4, T7, T9, T10	5
DE	T5, T9, T10	3

L2 — Frequent 2-itemsets (same as C2 since all passed)

2-itemset	Support
AB	4
AC	5
AD	3
AE	3
BC	5
BD	4
BE	4
CD	4
CE	5
DE	3

All candidates have support $\geq 3 \rightarrow \mathbf{L2}$ = these 10 itemsets.

Generate C3 and form L3

C3 — Candidate 3-itemsets (all 3-combinations) with TIDs & support

L3 — Frequent 3-itemsets

We list all possible 3-item combos of {A,B,C,D,E} and their supports.

Candidate	TIDs	Support	Frequent?
ABC	T1, T4, T9	3	Yes
ABD	T6, T9	2	No
ABE	T4, T9	2	No
ACD	T2, T9	2	No
ACE	T4, T7, T9	3	Yes
ADE	T9	1	No
BCD	T8, T9	2	No
BCE	T3, T4, T9	3	Yes
BDE	T5, T9	2	No
CDE	T9, T10	2	No

3-itemset	Support
ABC	3
ACE	3
BCE	3

So **L3 = {ABC (3), ACE (3), BCE (3)}**

Generate C4 and form L4

C4 — Candidate 4-itemsets (attempted joins & pruning)

- Potential 4-itemset from L3 would be ABCE (union of ABC and BCE or ABC and ACE).
- Check its 3-subsets: {ABC} (freq), {ABE} (not freq), {ACE} (freq), {BCE} (freq).
- Because subset ABE is not frequent, ABCE is pruned by Apriori property.

Therefore:

Candidate 4-itemset	Reason
ABCE	pruned — has non-frequent 3-subset (ABE)

No valid C4 $\rightarrow$ L4 = $\emptyset$. Algorithm ends.

Output:

For the given input transaction data and minsup =3, Apriori produced **frequent itemsets**: L1, L2 & L3. These itemsets are input for rule generation.

IMPORTANT

Association rules come **only from itemsets of size ≥ 2** (L2 and L3).

We will generate rules for:

- **All 2-itemsets** (rules like $A \rightarrow B$, $B \rightarrow A$)
- **All 3-itemsets** (rules like $AB \rightarrow C$, $A \rightarrow BC$, etc.)

And compute:

- **Support**
- **Confidence**
- **Lift**

SUPPORT & CONFIDENCE FORMULAS (We have 10 transactions)

- **Support($X \rightarrow Y$)** = support(XUY) / total transactions
- **Confidence($X \rightarrow Y$)** = support(XUY) / support(X)
- **Lift($X \rightarrow Y$)** = confidence($X \rightarrow Y$) / support(Y)

1. Association Rules from 2-Itemsets (L2)

For each 2-itemset XY, we get two rules:

- $X \rightarrow Y$
- $Y \rightarrow X$
- Total rules from L2 = $10 \times 2 = \mathbf{20 \text{ rules}}$

Table 1: Rules From 2-Itemsets

Rule	Support (XUY)	Conf($X \rightarrow Y$)	Lift
$A \rightarrow B$	4/10	$4/6 = 0.667$	$0.667 / 0.7 = \mathbf{0.953}$
$B \rightarrow A$	4/10	$4/7 = 0.571$	$0.571 / 0.6 = \mathbf{0.951}$
$A \rightarrow C$	5/10	$5/6 = 0.833$	$0.833 / 0.8 = \mathbf{1.041}$
$C \rightarrow A$	5/10	$5/8 = 0.625$	$0.625 / 0.6 = \mathbf{1.041}$
$A \rightarrow D$	3/10	$3/6 = 0.50$	$0.50 / 0.6 = \mathbf{0.833}$
$D \rightarrow A$	3/10	$3/6 = 0.50$	$0.50 / 0.6 = \mathbf{0.833}$
$A \rightarrow E$	3/10	$3/6 = 0.50$	$0.50 / 0.6 = \mathbf{0.833}$
$E \rightarrow A$	3/10	$3/6 = 0.50$	$0.50 / 0.6 = \mathbf{0.833}$
$B \rightarrow C$	5/10	$5/7 = 0.714$	$0.714 / 0.8 = \mathbf{0.892}$
$C \rightarrow B$	5/10	$5/8 = 0.625$	$0.625 / 0.7 = \mathbf{0.893}$
$B \rightarrow D$	4/10	$4/7 = 0.571$	$0.571 / 0.6 = \mathbf{0.951}$
$D \rightarrow B$	4/10	$4/6 = 0.667$	$0.667 / 0.7 = \mathbf{0.953}$
$B \rightarrow E$	4/10	$4/7 = 0.571$	$0.571 / 0.6 = \mathbf{0.951}$
$E \rightarrow B$	4/10	$4/6 = 0.667$	$0.667 / 0.7 = \mathbf{0.953}$
$C \rightarrow D$	4/10	$4/8 = 0.50$	$0.50 / 0.6 = \mathbf{0.833}$
$D \rightarrow C$	4/10	$4/6 = 0.667$	$0.667 / 0.8 = \mathbf{0.833}$
$C \rightarrow E$	5/10	$5/8 = 0.625$	$0.625 / 0.6 = \mathbf{1.041}$
$E \rightarrow C$	5/10	$5/6 = 0.833$	$0.833 / 0.8 = \mathbf{1.041}$
$D \rightarrow E$	3/10	$3/6 = 0.50$	$0.50 / 0.6 = \mathbf{0.833}$
$E \rightarrow D$	3/10	$3/6 = 0.50$	$0.50 / 0.6 = \mathbf{0.833}$

2. Association Rules from 3-Itemsets (L3) (ABC, ACE, BCE)

For each 3-itemset XYZ, rules are:

- $XY \rightarrow Z$
- $XZ \rightarrow Y$
- $YZ \rightarrow X$
- $X \rightarrow YZ$
- $Y \rightarrow XZ$
- $Z \rightarrow XY$
- Thus each 3-itemset gives **6 rules**.
We have 3 itemsets $\rightarrow$ 18 rules.

Table 2: Rules From L3 = ABC (support = 3)

Rule	Support	Confidence	Lift
$AB \rightarrow C$	3/10	$3/4 = 0.75$	$0.75 / 0.8 = \mathbf{0.937}$
$AC \rightarrow B$	3/10	$3/5 = 0.60$	$0.60 / 0.7 = \mathbf{0.857}$
$BC \rightarrow A$	3/10	$3/5 = 0.60$	$0.60 / 0.6 = \mathbf{1.00}$
$A \rightarrow BC$	3/10	$3/6 = 0.50$	$0.50 / 0.50 = \mathbf{1.00}$
$B \rightarrow AC$	3/10	$3/7 = 0.429$	$0.429 / 0.50 = \mathbf{0.858}$
$C \rightarrow AB$	3/10	$3/8 = 0.375$	$0.375 / 0.40 = \mathbf{0.937}$

Table 3: Rules From L3 = ACE (support = 3)

Rule	Support	Confidence	Lift
$AC \rightarrow E$	3/10	$3/5 = 0.60$	$0.60 / 0.6 = \mathbf{1.00}$
$AE \rightarrow C$	3/10	$3/3 = 1.00$	$1.00 / 0.8 = \mathbf{1.25}$
$CE \rightarrow A$	3/10	$3/5 = 0.60$	$0.60 / 0.6 = \mathbf{1.00}$
$A \rightarrow CE$	3/10	$3/6 = 0.50$	$0.50 / (5/10) = \mathbf{1.00}$
$C \rightarrow AE$	3/10	$3/8 = 0.375$	$0.375 / (3/10) = \mathbf{1.25}$
$E \rightarrow AC$	3/10	$3/6 = 0.50$	$0.50 / 0.50 = \mathbf{1.00}$

Table 4: Rules From L3 = BCE (support = 3)

Rule	Support	Confidence	Lift
$BC \rightarrow E$	3/10	$3/5 = 0.60$	$0.60 / 0.6 = \mathbf{1.00}$
$BE \rightarrow C$	3/10	$3/4 = 0.75$	$0.75 / 0.8 = \mathbf{0.937}$
$CE \rightarrow B$	3/10	$3/5 = 0.60$	$0.60 / 0.7 = \mathbf{0.857}$
$B \rightarrow CE$	3/10	$3/7 = 0.429$	$0.429 / 0.50 = \mathbf{0.858}$
$C \rightarrow BE$	3/10	$3/8 = 0.375$	$0.375 / 0.40 = \mathbf{0.937}$
$E \rightarrow BC$	3/10	$3/6 = 0.50$	$0.50 / 0.50 = \mathbf{1.00}$

FINAL OUTPUT SUMMARY

- ✓ With help of Apriori Algorithm, total frequent itemsets found: 18
- ✓ Total association rules generated with the frequent itemsets: 38
 - (20 rules from L2 + 18 rules from L3)
- ✓ Strongest rule (lift > 1):

AE → C

Confidence = 1.00

Lift = 1.25

Perfectly strong and meaningful.

Apriori Algorithm → ARM → MBA

Example, tying all three together:

Dataset: Customers buy Milk, Bread, Eggs, Butter.

Step 1: Apriori Algorithm

- Finds frequent itemsets: {Milk, Bread}, {Bread, Butter}, etc..

Step 2: Association Rule Mining

- Generates rules: Milk → Bread, Bread → Butter, etc..

Step 3: Market Basket Analysis

- Business uses rules to:
 - ✓ Put Milk and Bread closer on shelves
 - ✓ Give discount combo: Bread + Butter
 - ✓ Recommend Butter when customer adds Bread to cart

Let's Solve one example:

TID	Items
T1	Gaming Mouse, Mechanical Keyboard, Mousepad
T2	Headset, Gaming Mouse, Energy Drink
T3	Console Controller, Gaming Mouse, RGB Light Strip
T4	Mechanical Keyboard, Headset, Mousepad
T5	Gaming Mouse, Mechanical Keyboard, Headset, Mousepad
T6	Energy Drink, RGB Light Strip, Headset
T7	Gaming Mouse, Mousepad, Energy Drink

Why ethics matters in Data Mining

Data mining = finding patterns from large datasets.

But these patterns affect real people — their loans, health decisions, jobs, insurance, etc.

So, **ethics** means:

- Doing the right thing
- Not harming people
- Being fair to everyone
- Respecting people's privacy

Example:

If a bank uses data mining to approve loans, and the model is biased, many honest people may get rejected — this is unfair and unethical.

1. Ethical Issues Across the Data Mining Process

Data Collection Ethical Issues

Problem: Collecting too much data

- Companies collect unnecessary data “just in case.”
- Feels like spying.
- Increases privacy risks.

Example:

A mobile app for torchlight asks for location and contact list — unnecessary, unethical.

Problem: No proper consent

- Users don't know what is being collected.

Example:

Websites tracking your browsing behaviour without telling you.

Preprocessing Ethical Issues

Problem: Removing data incorrectly → bias

- If many women or rural people are removed during cleaning, the model becomes biased.
- **Example:**
Deleting all “missing salary” entries. Women’s entries have more missing salary because they prefer not to disclose → unintentionally model discriminates.

Problem: Imputing values wrongly

- Simple methods like mean value may distort sensitive groups.
- **Example:**
If rural income data is missing, filling with city average makes rural population look richer than reality.

Model Building Ethical Issues

Problem: Algorithm learning bias

- Models learn from history. If history had discrimination, model repeats it.

Example:

A hiring model trained on past resumes.

Company mostly hired men previously → model learns “prefer male resumes.”

Problem: Black-box models

- Deep learning models are difficult to explain.
People don't know why they got rejected or approved.

Example:

A loan applicant gets rejected but bank cannot explain why → unfair.

Deployment Ethical Issues

Problem: No transparency

- Users are not told how decisions are made.

Problem: Automation without human review

- Fully automated systems may make mistakes.

Example:

Fraud detection marks someone's payment as suspicious → card gets blocked while travelling — no human review.

Post-Deployment Ethical Issues

Problem: Model drift

- Data changes but model stays same → becomes unfair later.

Problem: Using data for new purposes

- Using data for something that users never approved.

Example:

Fitness app using your weight/steps data to sell you insurance without your permission.

Ethical Issues in Specific Data Mining Techniques

Classification (Decision Trees, SVM, Neural Networks)

Issue: Classification can become biased

- Model may give different results to different groups.

Example:

Loan model rejects more women because training data had fewer women with loans.

Issue: Hidden discrimination

- Even if we remove gender, other attributes reveal it indirectly.

Example:

ZIP codes may indirectly indicate caste/class/race → bias returns.

Clustering (K-means, DBSCAN)

Issue: Clusters may reveal private info

- People are grouped based on behaviour, but the group itself may reveal secrets.

Example:

Clusters in medical data → “diabetic group”, “HIV-risk group”.

This information is very sensitive.

Issue: Labelling groups wrongly

- Once clusters are formed, companies give labels like:
- “Low-value customers”
- “High-risk customers”
- This creates discrimination.

Association Rule Mining (Apriori, FP-Growth)

Issue: Reveals secret patterns

- Association rules may expose private habits.

Example:

Rule: “People who buy pregnancy test kits also buy calcium tablets.”
This can reveal pregnancy without consent.

Issue: Inference attacks

- Even if data is anonymised, patterns reveal identity.

Example:

If a person buys gluten-free items often, retailers assume they have a medical condition.

Regression (Linear/Logistic Regression)

Issue: Predicting sensitive outcomes

- Predicting:
- Credit score using social media
- Insurance risk using zip code
- These predictions can be unfair.

Issue: Spurious correlations

- Model might think two things are related when they aren't.

Example:

People in low-income areas miss EMI more → model marks entire area as “bad customers” → unfair to individuals who pay regularly.

Outlier Detection

Issue: False positives

- Normal behaviour appears abnormal.

Example:

A farmer makes a big deposit once a year → system flags as fraud → account blocked.

Issue: Minority behaviour looks like anomaly

- Models assume majority behaviour is normal.

Example:

If most customers shop daily, a weekly shopper gets flagged.

Recommender Systems

Issue: Filter bubbles

- People see only similar content.
Limits knowledge.

Example:

If you watch political videos of one side, YouTube shows only that side.

Issue: Manipulation

- Platforms push addictive videos for profit.

Example:

TikTok pushing endless short videos — reduces attention span.

Core Ethical Principles

Privacy

- Only collect needed data.
Don't misuse or overshare it.

Example:

A rideshare app should not check your gallery

Transparency

- People should know:
- What data is collected
- Why it is collected
- How decisions are made

Accountability

- Someone must take responsibility when the AI makes a mistake.

Fairness

- No discrimination based on:
- Gender
- Religion
- Caste
- Race
- Income

Do No Harm

- Model should not cause:
- Emotional harm
- Financial harm
- Social harm

Example:

Rejecting a loan without a reason can harm a person's life goals.

Consent

- User must agree before data is used.

Example:

Your health app cannot sell your steps data to insurance companies without asking.

Real-World Case Studies

Case 1: Cambridge Analytica

- Facebook likes → political personality prediction
- Used to manipulate election ads
- People didn't know their data was used
- Biggest ethics scandal

Lesson:

Always get consent.

Amazon Hiring AI

- AI shortlisted resumes
- AI preferred men
- Because historical data was male-dominated
- Amazon discontinued the system

Lesson:

Historical bias → model bias.

Legal & Regulatory Frameworks

GDPR (Europe) – Simple Explanation

- Tell users what data you collect
- Get consent
- Delete data if user asks (“Right to be forgotten”)
- Explain automated decisions

India – DPDP Act 2023

- Consent-based data usage
- Data must be used only for the specific purpose
- Heavy penalties for misuse

Best Practices

- Collect minimum data
- Remove sensitive attributes
- Test models for bias
- Explain your models
- Allow user to opt out
- Continuously audit models
- Use privacy techniques (anonymisation, differential privacy)

THANK YOU